



DALAS Project Report

What Makes a TED Talk Viral ?

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Abstract

TED Talks have become a primary vehicle for spreading ideas globally, yet only a fraction achieve viral status. This study examines whether viral success can be predicted using intrinsic talk characteristics available before publication. We compiled a dataset of 6,396 TED talks (2006–2025) and enriched with speaker information from Wikipedia. Moreover, we defined virality as the top 10% normalized daily viewership. Through exploratory data analysis, we found that viral talks are characterized by moderate length (30-60 minutes), greater linguistic depth, and high interactivity as well as prior TED platform experience. Our final classification model, built using XGBoost, achieves **73% recall** and a **PR-AUC of 0.21**. While precise binary prediction remains difficult due to the inherent randomness of online sharing, the model identifies most viral talks and offers a **2.9× efficiency gain** over random selection. This demonstrates that virality is influenced by intrinsic characteristics. Nonetheless they can not account for online user engagement and social media dynamics.

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Chapter 1

Introduction

1.1 Context

TED Conferences, LLC is a nonprofit media organization renowned for distributing impactful talks online. Originally focused on Technology, Entertainment, Design (TED) and has now broadened its topics by including education, politics, health, culture, science, technology and personal development and more. Nowadays, TED Talks are one of the most influential way to spread knowledge globally. Over the past years, TED Talks have had a significant cultural impact in the public discourse by popularizing or simplifying complex ideas, influencing education, innovation and social change on a global scale. Mostly delivered in English and translated in many languages, they reach a worldwide audience. Thousands of talks have been viewed online by highly diverse audiences, with some reaching tens of millions of views while others remain relatively unnoticed despite addressing similar topics.

1.2 Problem Statement

While a small number of talks achieve global recognition and continue to attract viewers years after their release, the majority remain confined to limited audiences. This unequal visibility is a striking phenomenon, particularly given the shared platform, format, and distribution mechanisms of TED Talks. However, they can have different durations, topics, or even settings. Identifying the characteristics that contribute to a talk’s success can offer valuable insights into how ideas can gain traction.

Most existing studies approach this issue by analyzing engagement indicators observed after publication, such as view counts, likes, or shares. While informative, these metrics provide limited insight into whether a talk’s success could have been anticipated beforehand. Therefore, the objective of this project is to predict the viral potential of a TED Talk solely based on its intrinsic characteristics available at or before the time of publication.

Accordingly, our core objective is to identify and quantify the factors that contribute to a TED Talk achieving “viral” status. Using a dataset comprising several thousand talks, we aim to understand why only a small fraction captures widespread and sustained public attention. More specifically, we seek to assess the extent to which thematic content, the speaker’s profile, and the publication context influence a talk’s success.

To address this question, after creating a dataset of all existing TED Talks and their speakers, we analyzed which features are correlated with high viewership and long-term public interest. Building on this analysis, we developed a predictive model that classifies TED Talks as “viral” or “non-viral”, relying exclusively on intrinsic characteristics rather than *a posteriori* engagement metrics.

1.3 Our Approach

1.3.1 Methodological Plan

The central question guiding this project is: what makes a TED Talk go viral and can the viral outcome of a talk be reliably predicted using only information available prior to or at the time of release? To answer it, we followed the 4-step plan below:

1. **Data Collection:** To construct a comprehensive dataset of TED Talks and their speakers, we performed web scraping of all publicly available talks hosted on TED.com using the BeautifulSoup

library. For each talk, we collected all available attributes and metadata. In addition, we searched and scraped the Wikipedia page of every speaker, if found by our algorithm.

2. **Data Preprocessing and Feature Engineering:** This phase includes data cleaning and handling missing values. We then defined the target variable by constructing two classes: viral (1) and non-viral (0). A talk is labeled as viral if it belongs to the top 10% of talks according to a normalized viewership-per-day metric, allowing us to account for differences in publication dates. Furthermore, relevant additional features were engineered.
3. **Exploratory Data Analysis:** We conducted an exploratory data analysis (EDA). Through data visualization and statistical analysis, we investigated distributions, trends, and patterns between features and the target variable.
4. **Predictive Modeling:** Finally, we developed predictive models for the binary classification. We used Logistic Regression, Linear SVM, Random Forest, and Histogram Gradient Boosting.

1.3.2 Ethical and Legal Aspects

Web scraping for this project was done with heavy consideration of ethical and legal constraints. Web scraping activities on TED.com as well as Wikipedia were performed in compliance with the websites robots.txt file. Moreover, we used delays between scraping requests to avoid heavy load on their servers.

All collected data consist exclusively of factual information that is publicly available. While TED Talk transcripts are copyrighted materials containing the speakers' original expressions, their use in this project is strictly limited to academic research and analytical purposes.

Finally, the dataset only includes information about public figures and publicly accessible content. No private, sensitive, or personally identifiable data are collected, stored, or processed at any stage of the project.

Chapter 2

Data Acquisition and Methodology

2.1 Overview of Data Sources

To analyze the factors that contribute to the virality of a TED Talk, we collected data from multiple complementary sources. Our strategy focused on acquiring both quantitative metrics (views, duration) and qualitative content (transcripts, topics) to enable a multidimensional analysis.

2.1.1 Primary Data Sources: TED.com

The official TED website (<https://www.ted.com>) served as the primary source, providing the core dataset. Key attributes collected include:

- **Talk Metadata:** Title, speaker name, publication date, duration, and event context.
- **Engagement Metrics:** Total view count across official platforms.
- **Content Classification:** Topic tags and content categories.
- **Transcripts:** Full-text transcriptions in English.

2.1.2 Complementary Data Sources: Wikipedia

A central hypothesis of this study is that virality may be driven not just by the content of a talk, but by the pre-existing public recognition of the speaker ("Speaker Fame"). To test this, we enriched our dataset by scraping Wikipedia. This allowed us to distinguish between "Platform Fame" (TED performance) and "Real-World Fame" (public recognition).

2.2 Data Collection Methodology

2.2.1 Web Crawling Strategy

Our crawling process began by systematically discovering all talk URLs from the TED.com archive. We implemented a duration-based filtering approach to handle the volume of content.

Technical Implementation

We utilized Python libraries including `requests` and `BeautifulSoup`. To ensure ethical compliance:

- **Rate Limiting:** We implemented a 2-3 second delay between requests to respect server resources.
- **Error Handling:** We built robust exception handling for connection timeouts and malformed HTML.

2.2.2 Detailed Scraping Process

Metadata Extraction

For each talk URL, we extracted structured data through targeted HTML parsing:

1. **View Count:** Parsed from dynamically loaded JavaScript content using regex patterns.
2. **Publication Date:** Retrieved from structured JSON-LD data and standardized to ISO format.
3. **Topics:** Collected from tag lists to analyze thematic breadth.

Transcript Collection

Transcripts were located in dedicated containers. We performed cleaning to remove HTML artifacts and timestamps, ensuring a clean corpus for Natural Language Processing (NLP). Approximately 13% of talks (mostly non-English or musical performances) lacked transcripts and were flagged for filtering.

2.2.3 Speaker Information Enrichment

Wikipedia Matching

To analyze speaker-level authority, we cross-referenced every speaker with Wikipedia.

1. **Cleaning and Splitting Speaker Names:** We parsed speaker strings to separate multiple speakers into individual entries by standardizing delimiters.
2. **Creating an Expanded Dataset:** Each speaker of a talk was represented by a separate row in an expanded data frame, retaining the associated data.
3. **Aggregating Speaker Data:** For each speaker, we aggregated all associated talks to compute:
 - Total number of videos, their IDs and date of publication
 - Combined list of unique topics across all their talks
4. **Output Dataset:** The resulting speaker dataset contains one row per speaker with aggregated features.

Scraping followed these steps:

1. **Name Processing:** For speakers with three or more names, we attempted combinations of different names.
2. **Web Scraping:** Retrieved the HTML content of each speaker’s Wikipedia page using requests and parsed it with BeautifulSoup.
3. **Data Extraction:** For matched pages, we extracted the biography summary and metadata and infobox (occupation, birth date).

Validation of External Data

To verify the quality of our Wikipedia matching:

- We manually audited a random sample of 200 speakers.
- **Result:** The matching algorithm achieved a 81% accuracy rate. For correctly matched pages, the data extraction showed an accuracy exceeding 91%. The 38 errors were due to:
 - **Ambiguous or Common Names:** Some speakers’ pages with common names include their profession in the URL title (e.g., “Kevin Kelly (editor)”), which our simple name-matching algorithm failed to capture.
 - **Mononyms or Artistic Names:** Speakers known by a single name or pseudonym (e.g., “Rives”) were occasionally matched to generic or unrelated pages.
 - **Long or Multi-Part Names:** Speakers with lengthy names sometimes failed to match due to substring mismatches or Wikipedia’s naming conventions.

While not perfect, the matching accuracy was deemed sufficient for our analytical purposes, as speaker-level Wikipedia features were used as supplementary rather than primary predictors.

2.3 Data Quality and Validation

2.3.1 Dataset Composition

The final dataset consists of:

- **Total Records:** 7,352 TED talks collected from 2006 through December 2024.
- **Transcript Availability:** 6,396 talks (87%) had usable English transcripts.
- **Speaker Enrichment:** Among 5,674 speakers, 2,797 speakers (49.3%) were successfully matched to a Wikipedia page.

2.3.2 Data Storage

All data was structured into two tables (Talks and Speakers) as detailed below.

Table 2.1: Primary Dataset Structure (Talks)

Column	Type	Description
id	Integer	Unique talk identifier
title	String	Talk title
speaker	String	Speaker name(s)
duration	Integer	Duration in seconds
views	Integer	Total view count
published_at	Date	Publication date (YYYY-MM-DD)
topics	List	Comma-separated topic tags
transcript	Text	Full transcript text
video_context	String	Event name (e.g., TED2018)

Table 2.2: Secondary Dataset Structure (Speakers)

Column	Type	Description
speaker	String	Unique Speaker Name
has_wikipedia	Bool	1 if speaker has a Wikipedia page
occupation	String	Extracted from Wikipedia infobox
wiki_summary	Text	First paragraph of biography

2.4 Ethical and Legal Compliance

Our data collection adhered to strict ethical guidelines:

- **Compliance:** We respected the `robots.txt` directives of TED.com.
- **Privacy:** No personally identifiable information (PII) of private citizens was collected; all data pertains to public figures.
- **Fair Use:** Copyrighted transcripts were used solely for feature extraction (e.g., word count, sentiment) and were not republished in raw form.

2.5 Bias and Limitations

We identified inherent biases in the raw data that informed our preprocessing strategy:

- **Temporal Bias:** Older videos have had years to accumulate views compared to recent uploads. We addressed this in Chapter 3 by creating a time-normalized `popularity_score`.
- **Language Bias:** The majority of talks are in English. We filtered out non-English talks to ensure consistency in linguistic analysis.

Chapter 3

Data Preprocessing

3.1 Data Cleaning

As our data came from one main source, it did not require significant cleaning. We conducted an Initial Data Analysis (IDA) to identify missing values and outliers and to determine if certain entries had to be discarded.

3.1.1 Missing Data

- **Transcripts:** We removed the talks with missing transcripts (often musical performances or podcasts) 6,396 videos left
- **Language:** Missing in 986 videos because of a default in scraping. Among the other videos, 27 are not in English. After verification, we noticed that they are all translated with an English title and transcript (or were removed as they did not have a transcript). Though we hoped to remove the Non-English videos as they do not necessarily reach the same audience, their English translation allows for minimal language bias. We then dropped the language column.

3.1.2 Outliers

We analyzed outliers in terms of duration and view count.

- **Duration:** While the standard TED format is 18 minutes, outliers exist ranging from 3 minutes to over 60 minutes. We chose to **keep** these outliers, as "extreme duration" (very short or long) is a valid feature that might influence virality.
- **Views:** The view distribution is highly skewed (long-tail), making absolute thresholds (e.g., "1 million views") unreliable due to platform growth over time. We therefore utilized **rank-based thresholding**, defining "Viral" as the Top 10% of talks by popularity score.

This threshold was chosen for two reasons: 1. **Robustness:** Relative ranking neutralizes the temporal inflation of view counts (a talk from 2006 vs 2024). 2. **Learnability:** The top 10% isolates the "breakout" tier of content while preserving enough positive samples ($N \approx 640$) for the model to learn statistical patterns, whereas a stricter threshold (e.g., 1%) would be too sparse for training.

3.2 Feature Engineering

A critical challenge in analyzing long-term web data is Temporal Bias: older videos have had significantly more time to accumulate views than recent uploads. To address this, we shifted our target variable from raw views to a normalized "Popularity Score" (views per day).

3.2.1 Main Data Features

We derived several features to capture the intrinsic quality and context of the talk:

- **popularity_score:** The primary metric for normalization, calculated as $\frac{\text{Total Views}}{\text{Days Since Published}}$. This allows us to compare a 2006 talk fairly with a 2024 talk.
- **is_extreme_duration:** A binary flag (0 or 1) indicating if a talk is significantly outside the norm (e.g., < 3 minutes or > 30 minutes).

- **days_since_published:** The age of the video in days, used as the denominator for the popularity score.
- **is_viral:** The target variable. A binary label set to **1** if the talk's **popularity_score** is in the top 10% of the dataset, and **0** otherwise.
- **is_viral_top5:** An auxiliary label for the top 5% of talks, used for stress-testing the model on "super-viral" hits.
- **content_category:** A broad categorization (e.g., Technology, Science, Culture) derived from the specific topic tags to reduce unique values ($334 \rightarrow 28$).

3.2.2 Speaker Data Features

The Speaker Data Features aim to help us understand if speakers that have already gone viral are more likely to go viral again than others. It helps us quantify "Reputation" versus "Content."

- **average_video_popularity_score:** The mean popularity score of all previous videos by the speaker.
- **viral_rate:** The proportion of viral videos among all the speaker's TED talks. We treat this as a proxy for the speaker's platform authority.
- **viral_rate_before_video:** To prevent data leakage, this feature calculates the proportion of viral videos among the speaker's talks released strictly before the current video. It is stored as a list of tuples (video date, **viral_rate_before_video**).

In order to join both datasets, we turned the speaker column into a list of speaker(s) in the main dataframe. Then, for each video, we added the average viral rate of its speaker(s), along with the max and min viral rate of all the video's speaker(s). We applied the same logic for the **viral_rate_before_video** to ensure the model only learns from past events.

Chapter 4

Exploratory Data Analysis (EDA)

4.1 Executive Summary

This report analyzes a dataset of TED Talks (2006-2025) to identify the distinguishing characteristics of "Viral" content. Contrary to the trend toward short-form media, virality shows a positive correlation with content depth. While the standard TED format is 18 minutes, talks in the "Very Long" category (30-60 minutes)** achieved the highest viral rates (22%), though they represent a small fraction of the total. Viral talks are also characterized by specific linguistic patterns: higher use of questions and generally positive sentiment.

4.2 Introduction and Data Overview

The objective of this analysis is to determine the metadata and content features that serve as reliable predictors for virality. The dataset was segmented into two cohorts: "Viral" and "Non-Viral" based on discretised view count thresholds. We explored duration, sentiment, linguistic style and more.

4.3 Temporal Analysis (Macro Trends)

Before analyzing specific features, we examined the evolution of the content ecosystem over the last two decades.

As illustrated in Figure 4.1, there is a clear macro-trend of decreasing talk length. However, coincident with this shift is a decline in average views per talk (Bottom Left). This suggests that while the platform has shifted toward shorter content, this strategy may not align with audience demand for high-impact viral content.

4.4 Feature Analysis

The strongest differentiators observed in our analysis are to the volume and depth of the content. We analyzed both talk duration and multiple measures of linguistic density in order to test what we refer to as the "depth hypothesis".

4.4.1 Duration and Virality

Despite the temporal trend toward brevity noted in the previous section, the data indicates a slight "Longer is Better" phenomenon for virality. As seen in Figure 4.2, the viral rate increases linearly with duration category. Talks categorized as **Short** (≤ 6 min) have a viral rate of $< 6\%$, while those categorized as **Very Long** (> 18 min) approach 18%. Still, 82% of long talks are not viral. Therefore, this finding is to be handled with care.

The figure above shows a more accurate division of duration category, which shows that being moderately long appears safer than being exceptionally long as 30-60 minutes videos have a viral rate of 22% whereas over 60 minute long videos have a viral rate of 11%.

4.4.2 Linguistic Complexity

Linguistic features show statistically significant differences ($p < 0.001$).

Figure 4.4 shows that viral talks contain significantly higher median *Word Counts* and *Sentence Counts*. This reinforces the hypothesis that virality in this domain is driven by comprehensive deep-dives rather



Figure 4.1: **Temporal Analysis.** (Top Right) Production volume peaked circa 2017–2019. (Bottom Right) Average talk duration has steadily declined from ≈ 18 minutes in 2006 to ≈ 11 minutes in 2024.

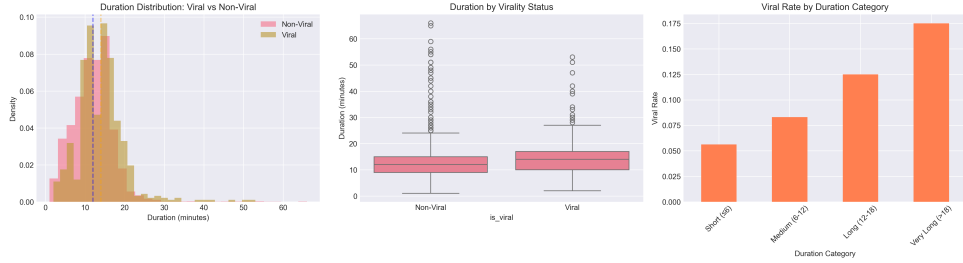


Figure 4.2: **Duration Analysis.** The bar chart (Right) shows that “Very Long” talks (> 18 mins) achieve a viral rate of nearly 18%, significantly outperforming “Short” talks.

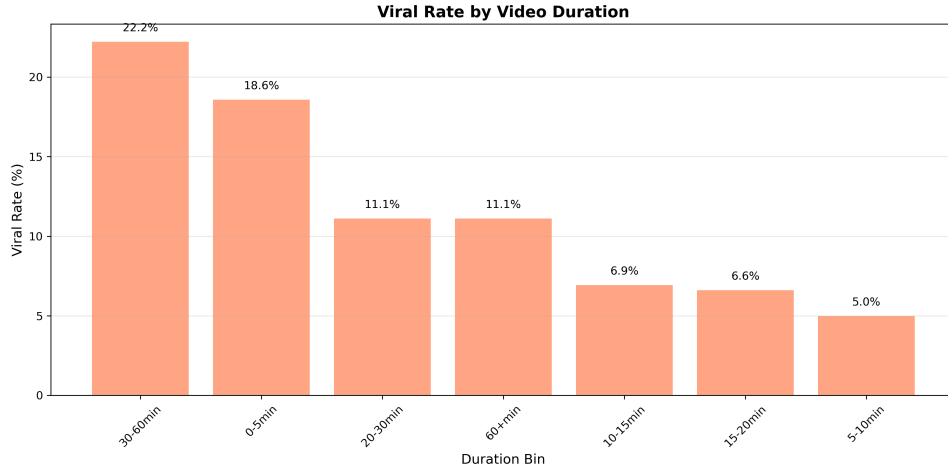


Figure 4.3: **Viral Rate by duration Bar Chart**

than high-level summaries. Notably, *Title Length* was found to be statistically insignificant, suggesting that optimizing titles for character count yields no measurable benefit.

4.4.3 Overlap analysis of categorical variables

To assess whether categorical variables could discriminate between viral and non-viral videos, we computed the Jaccard overlap between the sets of categories observed in each class. The Jaccard index measures the proportion of shared categories between two sets, with values close to 1 indicating a strong overlap and values close to 0 suggesting potential discriminative power.

The results are reported below:

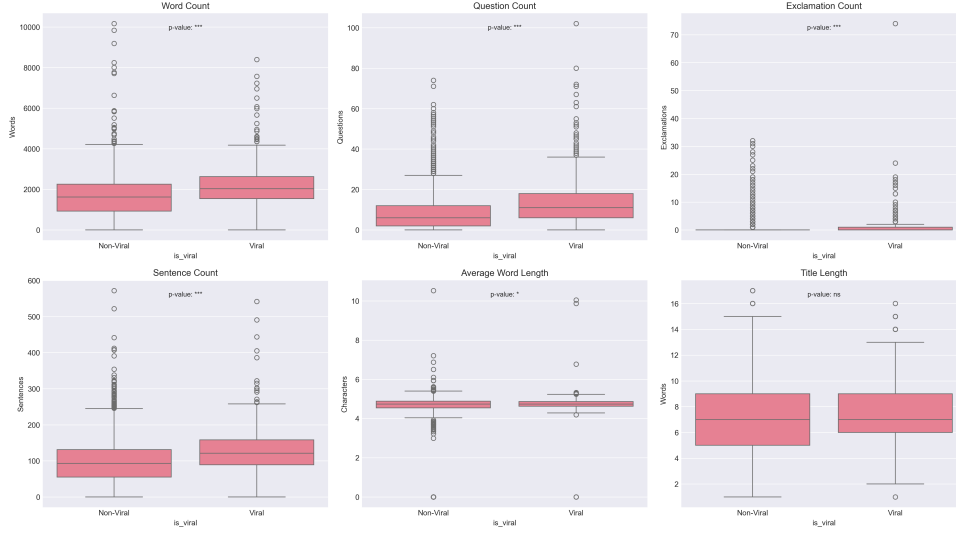


Figure 4.4: **Content Features.** Viral talks exhibit significantly higher word and sentence counts compared to non-viral talks. Note that Title Length differences are non-significant (ns).

Variable	Jaccard overlap
Type	1.00
Video context	0.16
Content category	1.00
Topics	0.85

The variables *type* and *content_category* exhibit a complete overlap between viral and non-viral videos, indicating that these attributes are not discriminative for virality. Similarly, the high overlap observed for *topics* suggests that viral and non-viral content largely share the same thematic space.

In contrast, the variable *video_context* shows a substantially lower Jaccard overlap (0.16), indicating that several contexts appear predominantly in either viral or non-viral videos.

4.5 Engagement and Sentiment Analysis

Beyond length, the tone and structure of the content play influential roles in audience retention and sharing.

4.5.1 Sentiment Polarity

The analysis reveals a “Positivity Bias.” As illustrated in Figure 4.5, talks classified as **Positive** have the highest viral rate, followed by Neutral and Negative. However, the overlapping histograms suggest that while positivity is a contributing factor, it is not a sole predictor; high subjectivity or polarity alone does not guarantee virality.

4.5.2 Audience Engagement (Questions)

A critical finding in the linguistic structure (referencing Figure 4.4, top middle) is the usage of questions. The **Question Count** is significantly higher for viral talks ($p < 0.001$). This indicates that successful speakers actively use rhetorical devices and direct questions to hook the audience, creating an interactive mental environment even in a one-way medium.

4.5.3 Expressivity

Exclamation marks are more frequent in viral talks, reflecting a more expressive and emotional tone, although this effect is driven by a subset of talks, as shown by the low median (referencing Figure 4.4, top right).

4.6 Topic Diversity

Here, we analyzed the number of topic tags associated with each talk as a proxy for interdisciplinary breadth.

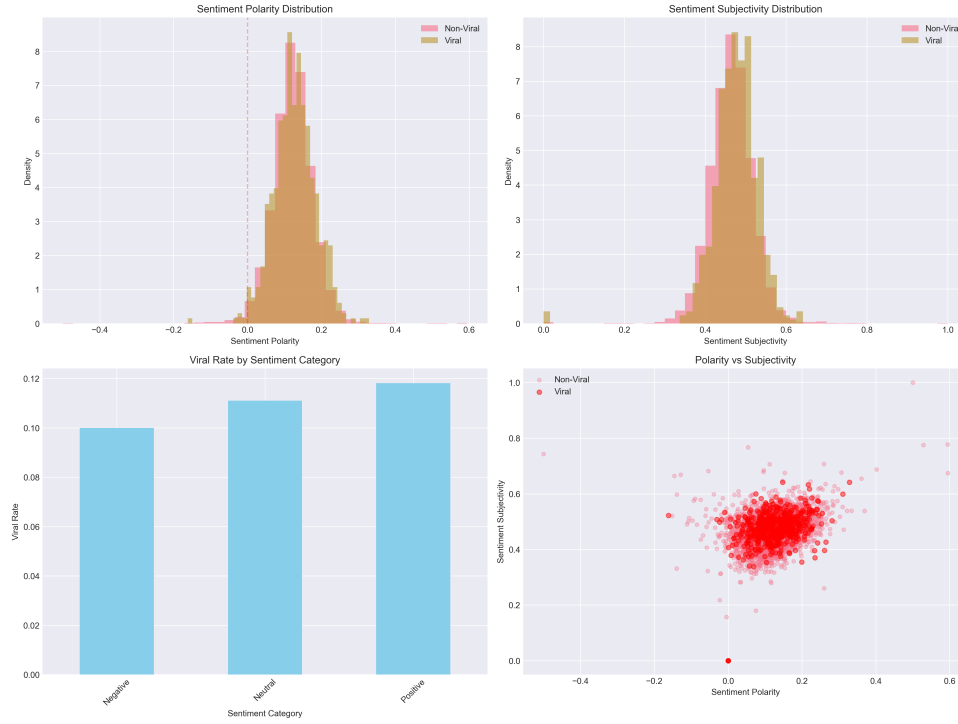


Figure 4.5: **Sentiment Analysis.** Positive talks (Bottom Left) show a marginally higher viral rate ($\approx 12\%$) compared to neutral or negative talks.



Figure 4.6: **Topic Diversity.** The line chart (Right) indicates an upward trend in viral rate as the number of topics increases toward 15.

The data suggests an “Interdisciplinary Sweet Spot.” Talks that bridge multiple disciplines (tagged with 10–15 topics) perform better than narrow, single-subject presentations. This correlates with the *Duration* findings, implying that viral talks are those that take the time to connect a core idea to a broader web of subjects (e.g., Technology + Design + Psychology). While the average number of topics is similar for both groups (around 7), viral talks leverage greater thematic breadth.

There are 334 topics in TED. We categorized them into 28 categories and tried to see if topic “cohesion” mattered to check if there was indeed an interdisciplinary sweet spot. We notice that the distribution is similar. Topic cohesion is small for most videos. Broad subjects perform better. However, the distribution coincides with non-viral talks distribution. Density is slightly higher for the viral subset (1.8 vs 3.5 for topic cohesion = 0.15). Better performance can be explained by the fact that the more tags there are, the most likely the video is to come up during search: it appears in more categories.

4.7 Readability

Here, we noticed that readability alone does not appear to be a decisive factor in predicting virality. Nonetheless, talking speed can matter.

As shown in Figure 4.8, most talks exhibit relatively high readability according to the Flesch Reading Ease score computed on the transcripts. Both viral and non-viral talks have similar average scores, with means around 50. While the central tendencies largely overlap, non-viral talks display greater variability, including several low-readability outliers. Although talks with very low readability scores do not appear

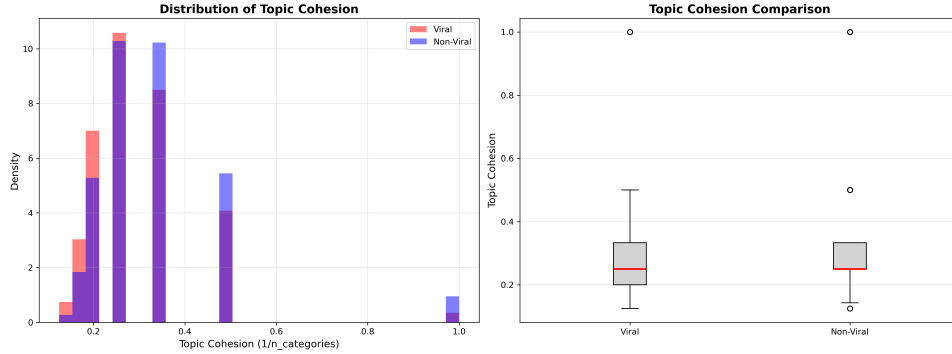


Figure 4.7: Topic Cohesion.

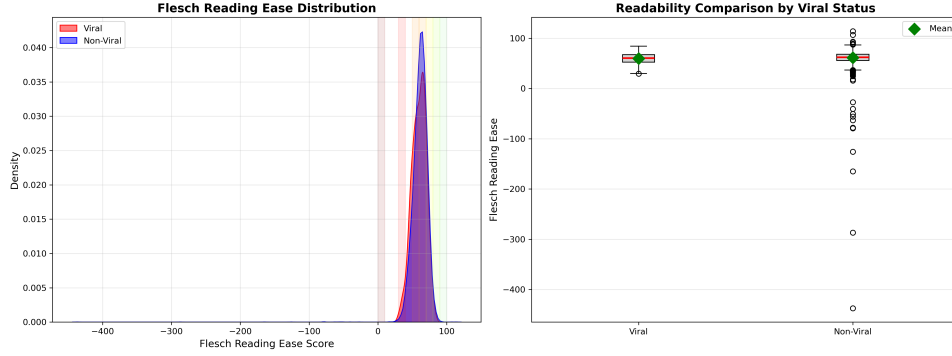


Figure 4.8: Flesch Reading Ease Comparison

among the viral subset, readability alone does not appear to be a decisive factor in predicting virality.

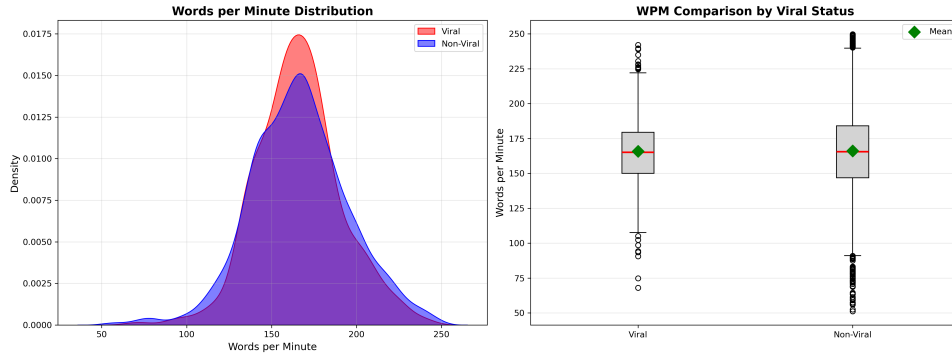


Figure 4.9: Words per Minute Comparison

In Figure 4.9, we observe that viral talks show a distribution shifted toward higher Words Per Minute (WPM), with a peak around 150–170 WPM: a dynamic yet comprehensible pace. They avoid slower rhythms and exhibit a more consistent speaking speed compared to non-viral talks, which show greater variance.

4.7.1 Speaker Feature Analysis

Among the top 20 speakers ranked by the average popularity score of their videos, half lack a Wikipedia page and only have one video, indicating that prior public recognition is not a prerequisite for video virality.

We checked further if whether the Wikipedia was found correlated with the viral rate of the speaker. Speakers with no Wikipedia page have a higher average popularity score than speakers with a Wikipedia page. However, we note that some Wikipedia pages were not found despite existing.

The median viral rate for speakers, with or without a Wikipedia page, is nearly identical and close to zero. This indicates that public recognition is not a decisive factor for virality, which aligns with how TED content is typically consumed: viewers select talks by topic, not speaker. Furthermore, about 90% of speakers have delivered only one talk, showing that virality is rarely driven by a speaker's fame or prior experience. This skewed distribution suggests that virality is rarely driven by speaker recognition

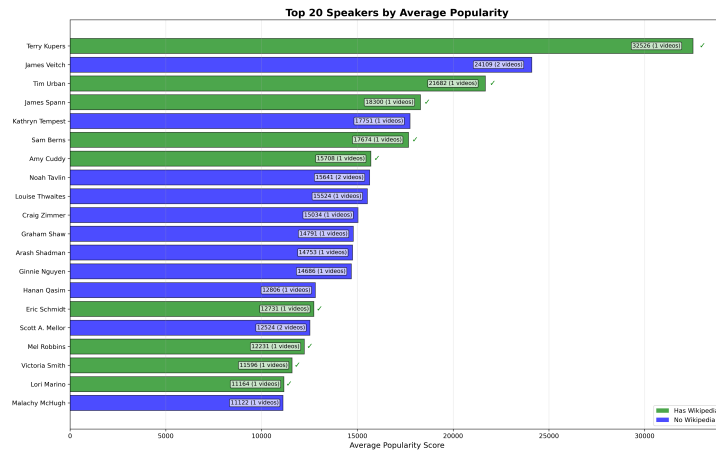


Figure 4.10: Top 20 Speakers

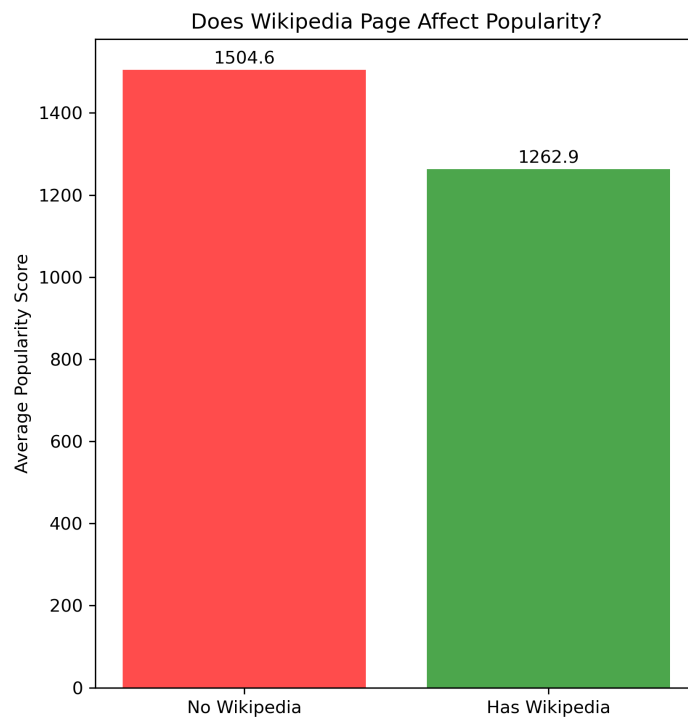


Figure 4.11: Wikipedia Analysis

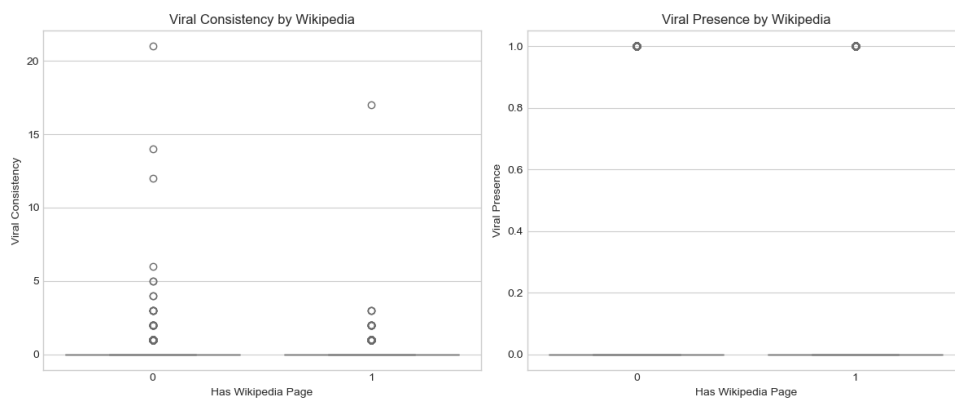


Figure 4.12: **Viral Consistency** Viral consistency is low for all speakers. It is dragged down by most speakers having a single talk that is not viral, for both speakers found on Wikipedia and those who were not.

or experience, as most viewers encounter new speakers.

Speakers with over seven talks achieve a median viral rate of 15% (IQR: 10~38%), compared to near-

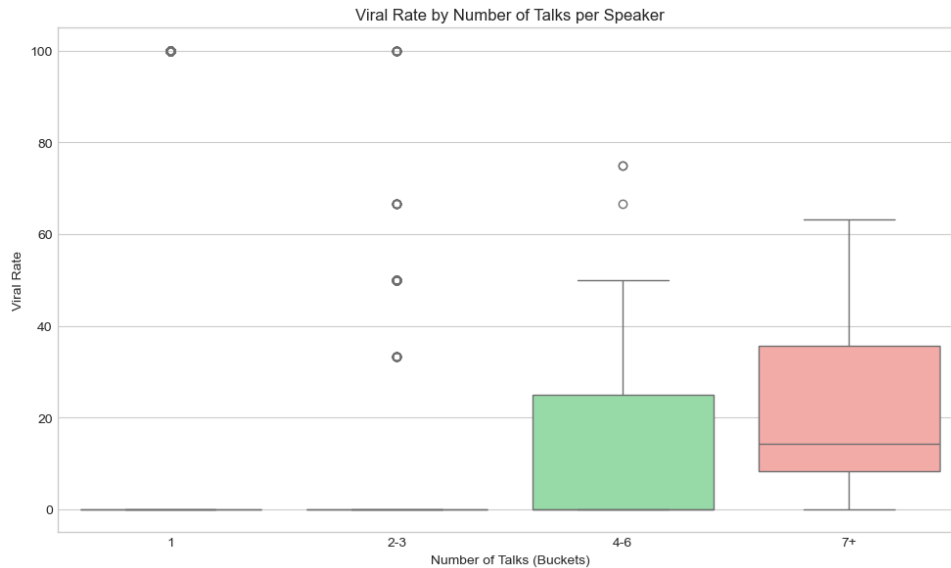


Figure 4.13: **Viral Rate by Speaker Experience.** Median viral rate increases modestly with the number of talks a speaker has given.

zero for most. This pattern suggests that having more talks slightly increases the likelihood of achieving virality, but the majority of speakers, even experienced ones, maintain low viral rates.

4.8 Conclusion and Strategic Recommendations

Based on the Exploratory Data Analysis, a higher probability viral TED talk is characterized by moderate length, content depth, positive tone, and audience engagement through questions. Speaker fame or prior recognition, as measured by Wikipedia presence, does not appear to be a decisive factor for virality. Having multiple talks slightly increases a speaker’s viral potential, but the majority of speakers—even those with several talks— maintain low viral rates.

- **Prioritize Depth:** Resist the pressure to cut content for brevity. Aim for talks around 30-60 minutes when possible, as moderate length outperforms extremely short or excessively long talks.
- **Engagement Architecture:** Actively incorporate questions into the script to increase audience participation and maintain attention.
- **Cross-Pollinate Topics:** Ensure the content touches on multiple interconnected themes or disciplines; talks with 10–15 topic tags tend to perform better.
- **Optimistic Framing:** Frame the narrative around positive outcomes or solutions to enhance shareability and retention.

Chapter 5

Predictive Modeling

This chapter details the machine learning methodology employed to predict the virality of TED Talks. We framed the problem as a binary classification task: detecting whether a talk will reach the top 10% of viewership based on metadata available prior to publication.

5.1 Data Preparation and Feature Engineering

5.1.1 Target Definition

We defined "Viral" as the top 10% of talks by popularity score. This threshold was chosen to isolate the minority class of high-performing talks given the **highly skewed distribution** of views (long-tail), where a small fraction of content accounts for the majority of engagement.

5.1.2 Sampling Strategy

Given the class imbalance (1:9 ratio), we implemented a **Hybrid Sampling Strategy** to improve model training:

- **BorderlineSMOTE**: To synthesize new "Viral" examples along the decision boundary and improve generalization.
- **RandomUnderSampler**: To balance the majority class and reduce computational noise.

Methodological Note: We utilized a **temporal split** (80/20) for model training. To mitigate potential leakage, all speaker history features (such as `viral_rate`) were calculated chronologically, using only data available *before* each specific talk's publication date.

5.1.3 Feature Engineering: Hypothesis Refinement

A key observation from our Exploratory Data Analysis (Chapter 4) was that external fame (Wikipedia presence) did not correlate with virality.

Consequently, we **refined our hypothesis**: while external fame appeared irrelevant, we postulated that a speaker's **internal track record** on the TED platform might be predictive. We therefore engineered specific historical features to capture this "Platform Reputation".

Comprehensive Feature Set

The final model was trained on the following set of features:

Category	Features Included
Speaker History	<code>viral_rate</code> (Historical viral %), <code>avg_views_prev</code>
Content Structure	<code>duration</code> , <code>word_count</code> , <code>sentence_count</code>
Linguistic Style	<code>exclamation_count</code> , <code>question_count</code> , <code>sentiment_polarity</code>
Temporal Context	<code>day_of_week</code> (One-Hot Encoded), <code>month</code> , <code>year</code>
Metadata	<code>type</code> (TED Stage vs TEDx), <code>tags_count</code>

Table 5.1: Complete list of features used for modeling. Note that all speaker history features were calculated chronologically to prevent data leakage.

5.2 Model Performance

We evaluated a Random Baseline, a Logistic Regression (Linear Baseline), and XGBoost (Tree-based ensemble).¹

5.2.1 Performance Analysis

Given the stochastic nature of virality, our objective was to maximize **Recall** (capturing the true positives) while maintaining an acceptable Precision.

Metric	Random Baseline	Logistic Reg.	XGBoost (Final)	Lift
PR-AUC	0.10	0.18	0.21	2.1x
Recall	10%	52%	73%	7.3x
Precision	10%	15%	18%	1.8x
F1-Score	0.10	0.23	0.29	2.9x

Table 5.2: Model Performance Comparison. The XGBoost model significantly outperforms the linear baseline, particularly in Recall.

Interpreting the F1-Score (0.29)

While an F1-score of 0.29 may appear low in absolute terms, it must be interpreted within the context of the problem complexity:

1. **High Difficulty:** Predicting viral phenomena is notoriously difficult due to external factors (social sharing, algorithms) that are invisible to the model. The random baseline F1 is only 0.10.
2. **Significant Improvement:** Our model provides a **2.9x improvement** over random guessing.
3. **Recall Trade-off:** By tuning the model to capture 73% of viral talks, we inevitably accept more false positives. However, for a screening use-case, minimizing false negatives (missed opportunities) is the priority.

5.2.2 Visualizing Utility: The Cumulative Gain Curve

While metrics like ROC-AUC assess theoretical ranking quality, they do not quantify practical efficiency. To demonstrate the **operational utility** of the model as a filtering tool, we prioritized the **Cumulative Gain Curve** (Figure 5.1).

¹We also tested Random Forest (F1=0.24) and Linear SVM (F1=0.21), but they underperformed XGBoost and were excluded from the final analysis. Final XGBoost hyperparameters: max_depth=6, learning_rate=0.05.

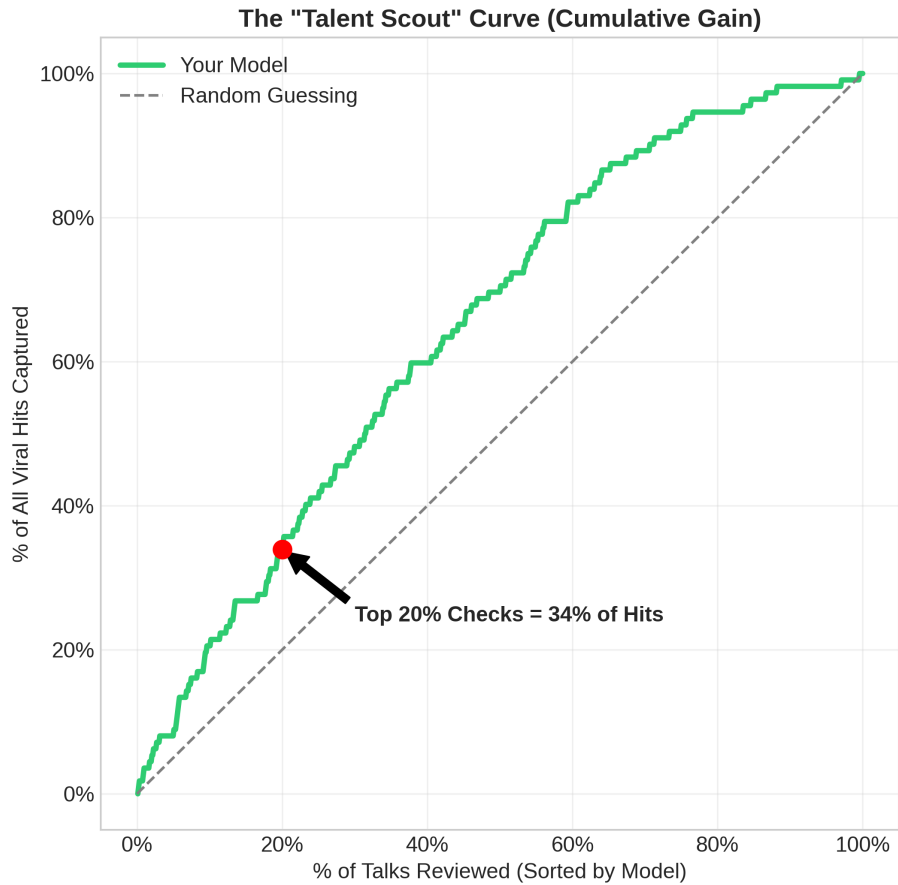


Figure 5.1: **Cumulative Gain Curve.** By reviewing the top 20% of talks ranked by the model, one can identify nearly 34% of all viral hits. This illustrates the model's effectiveness as a prioritization filter.

The curve shows that the model effectively concentrates viral candidates at the top of the list. At a 20% review rate, the model captures 34% of viral hits compared to 20% for random selection, representing a **1.7x efficiency gain** (Lift). This confirms that the model adds tangible value for a curator with limited time resources.

5.3 Feature Importance Analysis

The XGBoost feature importance analysis (Figure 5.2) provided insights into the drivers of prediction.

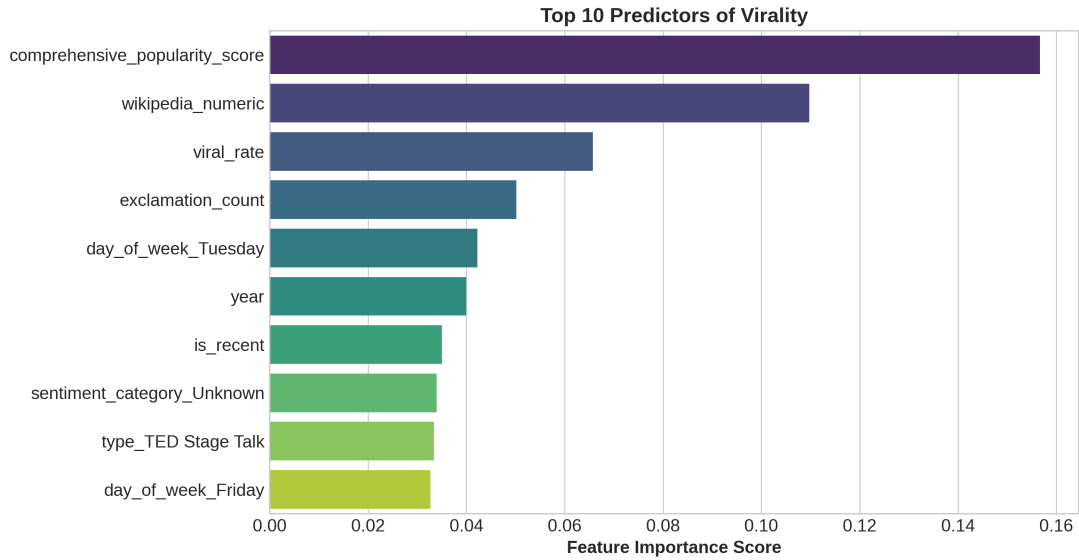


Figure 5.2: Top 10 Predictors of Virality. Emotional markers and timing are significant predictors.

1. **Emotional Engagement:** `exclamation_count` emerged as the strongest predictor. This suggests that high-energy delivery and emotional intensity are critical prerequisites for virality.
2. **Temporal Patterns:** `day_of_week_Friday` was a top predictor, indicating a correlation between release timing and consumption habits ("weekend effect").
3. **Platform Reputation:** `viral_rate` ranked 3rd. This validates our **refined hypothesis**: while external Wikipedia fame was irrelevant, a speaker's specific history on TED is a strong signal of future success.

5.4 Conclusion

The predictive modeling phase demonstrates that virality, while largely stochastic, contains a predictable signal. By focusing on internal speaker metrics and prioritizing recall, we developed a model that captures 73% of viral talks. While not deterministic, the model serves as an effective **prioritization filter**, distinguishing high-potential content from the noise with nearly 3x the accuracy of a random baseline.

Chapter 6

Limitations

Our analysis is subject to several limitations:

- **Temporal and Cultural Context:** The dataset spans over twenty years, during online video consumption, audience preferences and TED-s branding have significantly evolved, which our model does not account for.
- **Definition of Virality:** Virality was treated as a binary classification problem. This approach ignores the continuous nature of popularity.
- **Speaker Data Accessibility:** Speaker-level features relied on Wikipedia presence (49.3% match rate). While our manual audit of 200 speakers ($\approx 7\%$ of the cohort) estimated an accuracy of 81%, this implies that approximately 1 in 5 matched speakers may be incorrectly identified.
- **Platform Dynamics and Algorithmic Effects:** The analysis did not account for YouTube or TED.com recommendation algorithms or social media data such as shares on Twitter or public discourse on Reddit, although they can heavily influence a talk's reach.
- **Content Delivery and Non-Textual Features:** While transcripts were analyzed, visual aspects such as visual aids, stage presence, video production quality, and thumbnails were not included, though these likely impact viewer retention and sharing.
- **Absence of Early Engagement Metrics:** As per its premises, to avoid any leak from the future, the study excluded post-publication information. But including early indicators such as early views, likes, shares, or comments could improve predictive accuracy and help distinguish between immediate and long-term virality.
- **Class Imbalance and Validation Strategy:** The severe class imbalance constrained model performance. We used a temporal split (training on older talks, testing on recent ones) to better simulate real-world deployment.
- **Generalization After Deployment:** Deploying this model carries risks. Prioritizing high historical `viral_rate` could create feedback loops favoring established speakers, while the strong weight on `exclamation_count` might disadvantage speakers with calmer, analytical delivery styles.

Chapter 7

Conclusion

This project investigated whether the virality of a TED Talk can be predicted using only intrinsic characteristics available prior to publication. We found that viral talks tend to be moderately long, have greater content depth and emphasize on engaging viewers through questions, tone and dynamic talking speed. These findings suggest that content creators and TED speakers can improve the likelihood of success by optimizing content depth, engagement tactics, and publication timing, while acknowledging that no combination guarantees virality. Besides, the more videos they have on the platform, the more likely they are to go viral. However, these tendencies are neither universal nor sufficient on their own. The important overlap and small distribution differences observed between viral and non-viral talks suggest that virality emerges from an accumulation of weak signals rather by decisive factors. This was confirmed by our predictive modeling. It showed that simple, interpretable models like Logistic Regression can identify over half of viral talks with a 0.21 F1 Score, confirming that meaningful signal exists in intrinsic features. By prioritizing recall over precision, our final XGBoost model identified 73% of viral hits with a 0.29 F1 score. The modest performances (low F1 score) reflect the inherent unpredictability of audience engagement. Future research could build on this foundation by incorporating early engagement metrics and refining virality as a continuous or multi-class target. Furthermore, it would be interesting to investigate how much algorithmic suggestions amplify virality in digital media as algorithmic curation, cultural context and human curiosity converge.